FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

OPEN TEST
PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	A, B, C	19.	A, B, C, D	37.	A, B, C, D
2.	A, B	20.	B	38.	D
3.	A, C, D	21.	A, B, C, D	39.	A, B
4.	B, C	22.	B, C	40.	A, B, D
5.	B, C, D	23.	A, D	41.	A, C, D
6.	A, C	24.	A, B, D	42.	A, C
7.	A, C	25.	A, C	43.	B, D
8.	D	26.	C	44.	B
9.	C	27.	D	45.	B
10.	D	28.	B	46.	B
11.	B	29.	A	47.	D
12.	D	30.	B	48.	D
13.	A	31.	D	49.	C
14.	8	32.	0	50.	2
15.	3	33.	5	51.	4
16.	3	34.	3	52.	6
17.	8	35.	9.	53.	1
18.	3	36.	1	54.	3

Physics

PART – I

SECTION – A

1. $\frac{Q_{out}}{Q_{in}} = \frac{T_{out}}{T_{in}}$ and $Q_{out} = Q_{in} + W$

$\Rightarrow Q_{in} = W\left(\frac{T_L}{T_H - T_L}\right)$

$\frac{Q_{in}}{t} = P\left(\frac{T_L}{T_H - T_L}\right)$

2. Between faces LMNO and PQRS

$dR = \frac{\rho dr}{t\frac{\pi}{2}r}$

$R = \frac{2\rho}{t\pi}\ell n\frac{b}{a}$

Between faces LMRS and ONQP for an element at distance r from O.

$dR = \frac{\rho\frac{\pi}{2}r}{tdr}$

So, $\frac{1}{R} = \int\frac{1}{dR} = \int_a^b \frac{tdr}{\rho\frac{\pi}{2}r} = \frac{2t}{\rho\pi}\ell n\left(\frac{b}{a}\right)$

$R = \frac{\rho\pi}{2t\ell n\left(\frac{b}{a}\right)}$

3. $(i_1)_0 = \frac{50}{25} = 2$ amp

$(i_2)_0 = \frac{50}{5000} = 0.01$amp

$\tau_1 = 0.125$ sec

$\tau_2 = 0.125$ sec

$i_1 = 200\, i_2$

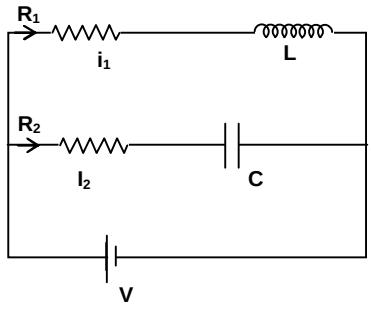
$2\left(1 - e^{\frac{-t}{0.125}}\right) = \left(0.01\, e^{\frac{-t}{0.125}}\right)200$

$t = \frac{\ell n2}{8}$

$4\left(\frac{q}{C}\right) = L\frac{di_1}{dt}$

$4\times 50\left(1 - e^{\frac{t}{0.125}}\right) = 50e^{-\frac{t}{0.125}}$

$t = \frac{1}{8}\ell n\left(\frac{5}{4}\right)$



4. $x^2 = \frac{2a}{\sqrt{3}}y$

$2x = \frac{2}{\sqrt{3}}a\frac{dy}{dx}$

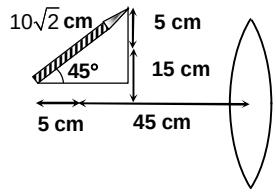
$\tan\theta = \sqrt{3}\frac{x}{a}$

$\theta = 60°$

$a = \frac{g\sin\theta}{1+\frac{I_{cm}}{mR^2}} = \frac{g}{\sqrt{3}}$

$f = \frac{mg\sin\theta}{1+\frac{mR^2}{I_{cm}}} = \frac{mg}{2\sqrt{3}}$

5. Length of pencil is $10\sqrt{2}$ cm.



Position of (A) [–(45+5), (15 – 5) $\Rightarrow$ (–50, 10)

$\frac{1}{v} - \frac{1}{(-50)} = \frac{1}{20}$

v = 33.3 cm

magnification $\frac{h_i}{h_0} = \frac{-v}{u} \Rightarrow h_i = -h_0\frac{v}{u} = 10\frac{100/3}{50} = 6.6$ cm

Position of B (40 cm, 20 cm)

$\frac{1}{v} - \frac{1}{-40} = \frac{1}{20} \Rightarrow v = 40$ cm

magnification $h_i = 20 \times \frac{40}{40} = 20$ cm

Position of C (36 cm, 12 cm)

$\frac{1}{v} = \frac{1}{20} - \frac{1}{45} \Rightarrow v = 36$ cm

Magnification $h_i = 15\frac{36}{45} = 12$ cm

6. $v_0 = 4\sqrt{gR}$

$\vec{J} = \Delta\vec{P}$

$-\int T dt = 2M\left(v - 4\sqrt{gR}\right)$

$-\int T dt = 2M\left(R\omega - 4\sqrt{gR}\right)$ …(i)

$\left(\int T dt\right)R = MR^2\omega$ …(ii)

From (i) and (ii)

$\omega = \frac{8}{3}\sqrt{\frac{g}{R}}$ and $v = \frac{8}{3}\sqrt{gR}$

7. $$x_{cm} = \frac{\int x dm}{\int dm} = \frac{\int_0^L x\lambda_0\left(1+\frac{x}{L}\right)}{\int_0^L \lambda_0\left(1+\frac{x}{L}\right)} = \frac{5L}{9}$$

$$m = \int dm = \frac{3\lambda_0 L}{2}$$

$$\left[\left(A\cdot\frac{L}{9}\right)\rho g\right]\frac{17}{10}L\sin\theta = F\cdot L\cos\theta$$

8-10. (1) For $\beta = \frac{D\lambda}{d}$ to be maximum, $D\lambda$ should be maxium and d should be minimum.

(2) In the Lloyd's experiment, there would not be any interference at point C.

(3) In option (A) and (C) no. Of fringes is much larger because the interference peattern is spread at the whole screen

In option (B) no. of fringes = $\frac{\text{Fringe pattern width}}{\text{Fringe width}}$

$$= \frac{\frac{d}{2}\left(9-\frac{3}{2}\right)d}{D\lambda} = \frac{15}{4}\frac{d^2}{D\lambda} = 240$$

In option (D) no. Of fringes = $\frac{\text{Fringe pattern width}}{\text{Fringe width}}$

$$= \frac{2(\mu-1)A(D-d)}{\frac{D\lambda}{2(\mu-1)Ad}} \approx \frac{4(\mu-1)^2A^2d}{\lambda} = 22$$

11-13. $\frac{1}{2}ma_{cm}^2t^2 + \frac{1}{2}I_{I_{cm}}\alpha^2t^2$

In each configuration kinetic energy of solid sphere will be maximum.

When F_{net} = 3F the kinetic energy of sphere will be maximum.

$$= \frac{14F^2}{m}t$$

When $F_1 = F_2 = F$

$$a_{cm} = \frac{2F}{m},\ R\alpha = \frac{FR^2}{I_{cm}}$$

When $F_2 = 2F_1 = 2F$

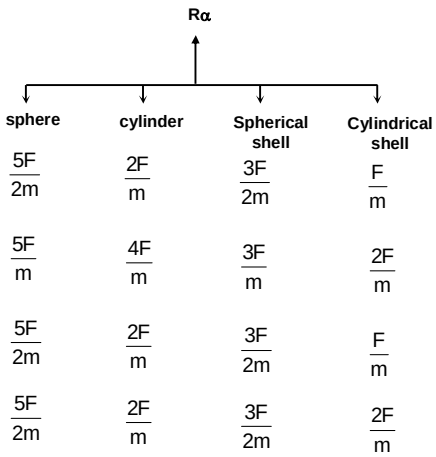
$$a_{cm} = \frac{3F}{m},\ R\alpha = \frac{2FR^2}{I_{cm}}$$

When $F_1 = 0$ and $F_2 = F$

$$a_{cm} = \frac{F}{m},\ R\alpha = \frac{FR^2}{I_{cm}}$$

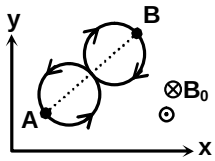
When $F_1 = \frac{3F_2}{2} = \frac{3F}{2}$

$$a_{cm} = \frac{5F}{2m},\ R\alpha = \frac{FR^2}{I_{cm}}$$



SECTION – C

14. $AB = \dfrac{4mv}{qB_0} = 8 \text{ metre}$



15. $\dfrac{\lambda_1}{\lambda_2} = \dfrac{\ell n2}{14 \text{ days}} \dfrac{28 \text{ days}}{\ell n2} = 2$

$\dfrac{A_{10}}{A_{20}} = \dfrac{\lambda_1 N_{10}}{\lambda_2 N_{20}} = 2 \times 4 = 8$

$\therefore\ A_{10} = 80$ m curie and $A_{20} = 10$ m curie

$A = 80e^{-\frac{\ell n2}{14} \times 80} + 10e^{-\frac{\ell n2}{28} \times 80}$

$= 80\ e^{-4} + 10\ e^{-2} = 3$ m curie

16. $C = \dfrac{2\pi ab\varepsilon_0 (k+1)}{(b-a)}$

$R = \dfrac{1}{\sigma} \dfrac{(b-a)}{\pi ab}$

$RC = \dfrac{2\pi ab\varepsilon_0 (k+1)}{(b-a)} \dfrac{(b-a)}{2\pi ab}$

$Q = Q_0 e^{-\frac{\pm\Sigma}{\varepsilon_0 (k+1)}}$

17. $m_1 a_1 = 6\pi\eta r_1 v_1$

$m_2 a_2 = 6\pi\eta r_2 (v_2 - v_1)$

$\dfrac{a_1}{a_2} = \dfrac{r_1 v_1 m_2}{r_2 (v_2 - v_1) m_1} = \dfrac{4}{3}$

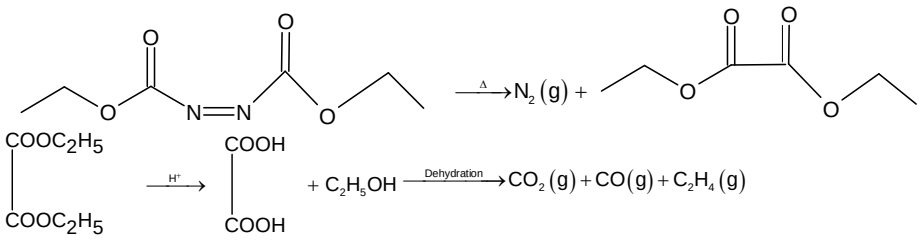
18. $\dfrac{v}{4\ell} = 412.5$ (3rd harmonic)

Chemistry

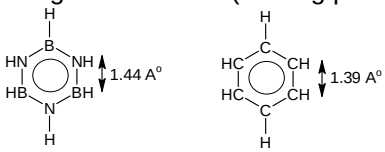
PART – II

SECTION – A

20. 6 geometrical isomers (Case I) + 40 optical isomers (Case II) = 46 space isomers.

21. 

22. Organic benzene being a more symmetrical molecule has higher melting point (6°C) than inorganic benzene (melting point = -58°C)



24. $MgCl_2.6H_2O \xrightarrow{\text{strong heating}} MgO\,(MgCl_2 \text{ reacts with water of crystallisation})$

$2Mg + \underset{(\text{Dryice})}{CO_2} \xrightarrow{\text{Combustion}} 2MgO + C$

$LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2$

Solution for the Q. No. 29 to 31.

Peptisation: Phenomenon in which a precipitate is passed into colloidal form by adsorption of common ions on its surface.

Delta formation: When river water (sol) meets sea water (electrolyte) coagulation occurs and delta is formed.

Tyndall effect: Scattering of light rays by suspended particles in a colloid is called Tyndall effect and sky looks blue due Tyndall effect.

Protection of colloids: Protective power of colloids is inversely related to Gold number.

SECTION – C

32. PV = nRT

$1 \times 8.21 = n \times 0.0821 \times 500$

n = 0.2 = mole of Argon

This implies that no decomposition of $NH_2COONH_4(s)$ has taken place at 227°C.

$\therefore K_p = 0$

33. $\underset{(\text{impure})}{Al_2O_3} + N_2(g)\uparrow + 3C \longrightarrow AlN + 3CO(g)\uparrow$

$\underset{(\text{impurity})}{SiO_2} + 2C \longrightarrow Si(v)\uparrow + 2CO(g)\uparrow$

$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3(g)\uparrow$

$2Al(OH)_3 \xrightarrow{\Delta} \underset{(\text{Pure})}{Al_2O_3} + 3H_2O(v)\uparrow$

35. $[Al^{+3}]\,[OH^-]^3 = 1 \times 10^{-18}$

$10^{-3}\,[OH^-]^3 = 1 \times 10^{-18}$

$[OH^-] = 10^{-5}$

pOH = 5

pH = 9

36. In given 100 ml 0.1 M NaCl solution,

No. of moles of Cl^- present = $\frac{100}{1000} \times 0.1 = 0.01$

And 1 Faraday of charge can oxidize 1 mole of Cl^- $\left(\because 2Cl^- \longrightarrow Cl_2 + 2e^-\right)$

$\therefore$ Given 0.05 F is not utilized in electrolysis
Only 0.01 F will be utilized in electrolysis

$\therefore$ From $2H_2O \xrightarrow{+2e^-} 2OH^- + H_2$

2 F of charge can produce 2 moles of OH^-

$\therefore$ 0.01 F will produce 0.01 mole of OH^-

$\therefore$ Molarity of OH^- in solution = $\frac{0.01 \times 1000}{100} = 0.1\,M$

$\therefore$ pOH = - log 10^{-1}
= 1

Mathematics PART – III

SECTION – A

37. $P = \dfrac{(1+2+3+.....+10)^2}{2} = \dfrac{1}{2}\left(\dfrac{10\times 11}{2}\right)^2 = \dfrac{55^2}{2}$

38. Function is continuous and differentiable at x = 0 and discontinuous everywhere else

39. $(BC)^T_{21} = BC_{12} = \dfrac{2a-2}{a+2+2a^2} = -1$

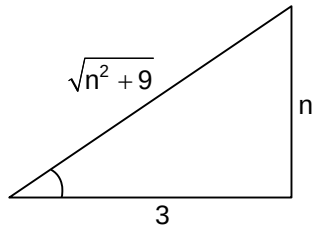
40. y = x + 1 is parallel to axis of the parabola so reflected ray will passes through its focus

41. Lines L_1 and L_2 are intersects at point (1, 0, 2)

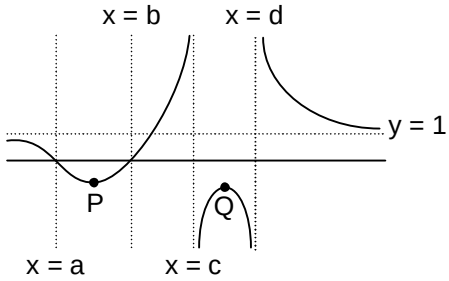
42. $\dfrac{e^x + e^{4/x}}{x} > e^x$ for $x \in (1, 2)$ and for option (A) put $x = \dfrac{4}{t}$

43. $T_r = \tan^{-1}\left(\dfrac{\frac{1}{3}}{1+\frac{r(r-1)}{9}}\right) = \tan^{-1}\left(\dfrac{r}{3}\right) - \tan^{-1}\left(\dfrac{r-1}{3}\right)$

$S_n = \sum_{r=1}^{n} T_r = \tan^{-1}\left(\dfrac{n}{3}\right)$



44.-46. We cannot be sure about the coordinates of P and Q as they depend on a, b, c and d



47.-49. $\int_0^x \sqrt{1-(f''(t))^2}\,dt = \int_0^x f'(t)\,dt \Rightarrow \sqrt{1-(f''(x))^2} = f'(x) \Rightarrow f''(x) = \pm\sqrt{1-(f'(x))^2}$

$\Rightarrow x = \pm\sin^{-1} f'(x) \Rightarrow f(x) = \pm(1-\cos x) = \pm 2\sin^2\dfrac{x}{2}$

$f(x) \le |f(x)| = 2\sin^2\dfrac{x}{2} < 2\cdot\dfrac{x^2}{4} = \dfrac{x^2}{2}$

$\int_0^1 f(x)\,dx \le \int_0^1 |f(x)|\,dx = \int_0^1 2\sin^2\dfrac{x}{2}\,dx < \dfrac{1}{2}\int_0^1 x^2\,dx = \dfrac{1}{6}$

SECTION – C

50. Two solutions exist in $(0, \pi)$

51. Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2} \Rightarrow \frac{z_1\bar{z}_2 + \bar{z}_1 z_2 + z_1 z_2 + \bar{z}_1\bar{z}_2}{|z_1 z_2|}$ = 2 cos $(\theta_1 - \theta_2)$ + 2 cos $(\theta_1 + \theta_2)$

52. Semi-perimeter of $\Delta S'PQ$ is 2a

53. Given a + b + c = 1 ..... (1)
9a + 3b + c = 7 ..... (2)
18 < 25a + 5b + c < 22 ..... (3)
$\Rightarrow$ From above (1), (2) and (3) 4 < 7a – b – c < 8
4 < 7a – b + a + b – 1 < 8
5 < 8a < 9
$\frac{5}{8} < a < \frac{9}{8}$ (a = 1)
a = 1, b = –1, c = 1
For question (2) $h(x) = \ln(x^2 - x + 1) - x$

54. $L = \frac{2}{3} \times e^2$